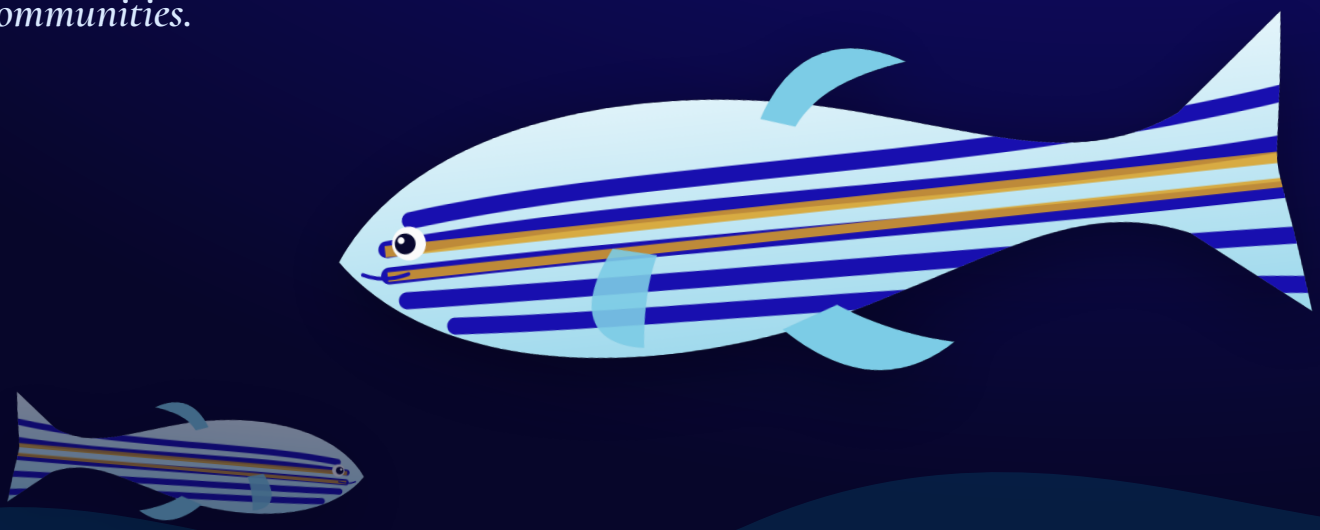
NZINC · NEPAL

Zebrafish in *Nepal*

A small native fish of Nepal's rivers and rice paddies, and a big idea for science, conservation and communities.



NEPAL ZEBRAFISH IN NATURE CENTER · BHARATPUR-7, CHITWAN

MEET THE SPECIES

A tiny fish with a Nepali home.

The zebrafish (*Danio rerio*) is native to the Indian subcontinent, with wild populations in the Ganges and Brahmaputra basins across Bangladesh, India and **Nepal**. Its pale blue body and bold horizontal stripes make it instantly recognisable, but in Nepal's lowland streams it is far more than a pretty fish.

~2.5 cm	~1 yr	6–38 °C	1981
Typical wild adult length	Typical lifespan in the wild	Temperature range tolerated	Used in biomedical research since

Where it lives

Zebrafish favour shallow, slow-moving water: stream margins, floodplains, ponds, ditches and irrigated fields. They are at home in farmland as well as in clear rivers, which ties their health closely to the way people use and care for water.

What it eats

An omnivore and a tireless forager, it feeds on zooplankton and aquatic insects, picks insects from the water surface, and readily eats mosquito larvae.

At a glance

- **Scientific name:** Danio rerio
- **Family:** Danionidae (minnows and danios)
- **Range:** Ganges and Brahmaputra basins
- **Habitat:** shallow, slow freshwater
- **Diet:** zooplankton, insects, larvae
- **Role in Nepal:** a native species worth knowing, studying and protecting

WHY IT MATTERS

Small fish. Global science.

Zebrafish are one of the world's most important research animals. Their embryos are transparent and develop outside the mother, so scientists can watch an animal take shape, cell by cell. They breed quickly and lay many eggs, which makes them ideal for studying how vertebrates develop, how genes work and how diseases begin.

> *"Before a native species can be protected, it has to be found, measured and understood."*

01

Science

A model organism for genetics, development, evolution and disease, used in laboratories around the world.

02

Conservation

An indicator of healthy freshwater. Knowing where native zebrafish live helps protect rivers, wetlands and the people who depend on them.

03

Nepal's own story

Wild populations here are part of the species' natural range, yet remain under-studied. Local research puts Nepal on the scientific map.

04

Communities

Healthy waters support farming, food and livelihoods, so research and rural development go hand in hand.

Four pillars that drive our impact.

Nepal Zebrafish in Nature Center (NZINC) is a non-profit, non-political, public-interest research organisation based in Chitwan. Its work brings research, innovation and community engagement together to address challenges in environmental sustainability, agriculture and food security.

I

Zebrafish & Aquatic Biodiversity

Studying zebrafish and other freshwater fish in their natural habitats through ecological studies, morphometric characterisation, habitat assessment and water quality analysis.

II

Molecular & Phenotypic Research

Advancing molecular diagnostics, genetic characterisation and phenotypic assessment to support discovery, disease identification and a better understanding of biological diversity.

III

Sustainable Agriculture & Livestock

Supporting crop protection, indigenous and improved livestock research and climate-resilient farming to strengthen Nepal's food security.

IV

Climate Change & Conservation

Studying environmental change and its effects on biodiversity and agriculture, and promoting evidence-based policy and sustainable management of natural resources.

Vision

A well-recognised, learning and innovative institution for community and humane development, working as a connector that helps people access the resources they have a right to.

PARTNERSHIPS & IMPACT

Better together.

25+	50+	21+	4+
Projects completed	Volunteers working with us	INGO partners	Government partners

Research and academic partners

Duke University, USA · University of Oregon, USA · Agriculture & Forestry University · Mid-West University · Far-West University · Bharatpur Municipality

Recent milestones

MoU with Duke University

A research partnership to advance work on native zebrafish, aquatic biodiversity and river ecosystem conservation in Nepal.

Agreement with Agriculture and Forestry University

A mutual agreement promoting collaborative research, academic exchange and scientific cooperation in zebrafish research, agriculture and related sciences.

Registered and affiliated

NZINC is legally registered with the District Administration Office, Chitwan, and affiliated with the Social Welfare Council of Nepal.

Community work

From wetland restoration and basin action planning on the Narayani River to sustainable agriculture programmes that improved incomes for 1,200 smallholder farmers in Nawalparasi and Chitwan, NZINC links river health to rural livelihoods.

GET INVOLVED

Everyone can help.

Volunteering, partnering or donating directly supports freshwater research, nature restoration and community resilience across Nepal.

Volunteer

Join field programmes in nature restoration, river basin research, outreach and livelihoods. Build practical experience alongside dedicated experts.

Partner

Collaborate on research, academic exchange or community programmes that protect Nepal's aquatic biodiversity.

Donate

Help fund zebrafish and aquatic research, field work and the communities who live beside Nepal's rivers.

Nepal Zebrafish in Nature Center (NZINC)

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